Reg. No.															

B.TECH / M.TECH. (Integrated) DEGREE EXAMINATION, DECEMBER 2024

First Semester

21CYB101J - CHEMISTRY

(For the candidates admitted during the academic year 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
1.	For a high spin d^4 octahedral complex the crystal field splitting energy will be A) $-0.6\ \Delta_o$ B) $-0.8\ \Delta_o$ C) $0.6\ \Delta_o$ D) $0.8\ \Delta_o$	1	3	1	1
2.	Which of the following is paramagnetic? [Z for Co= 27, Fe=26, Ni= 28] A) [Pt(NH3)3Cl]+ B) [Ni(CN)4]2− C) $[CoBr]_4^{2-}$ D) $[Co(NH_3)_6]^{3+}$	1	2	1	1
3.	The crystal field splitting energy for octahedral and tetrahedral complexes is related as A) $\Delta_t \approx 1/2\ \Delta o$ B) $\Delta o \approx 2\ \Delta t$ C) $\Delta o \approx 4/9\ \Delta t$ D) $\Delta_t \approx 4/9\ \Delta o$	1	2	1	1
4.	Suggest the hard acid from the following ions given: A) Cu^+ B) Mg^{2+} C) Cd^{2+} D) Hg^{2+}	1	2	1	1
5.	According to the convention, the Daniel cell is represented as _____ A) Zn l $ZnSO_4$ll Cu l $CuSO_4$, E = 1.09 volt B) $ZnSO_4$ l Zn ll $CuSO_4$ l Cu, E = 1.09 volt C) Zn l $ZnSO_4$ll $CuSO_4$ l Cu, E = 1.09 volt D) Zn l ZnSll $CuSO_4$ l Cu, E = 1.09 volt	1	2	2	2
6.	The following are state functions EXCEPT A) enthalpy (H) B) heat (q) C) internal energy(U) D) entropy (S)	1	2	2	2
7.	When zinc metal, is attached to the ship's hull made of iron, zinc corrode itself instead of the hull. The type of corrosion is an example of ---- A) Galvanic corrosion B) oxidation corrosion C) liquid metal corrosion D) pitting corrosion	1	3	2	2
8.	If the solubility product of magnesium hydroxide is 2.00×10^{-11} mol^3 dm^{-9} at 298 k, calculate its solubility in mol dm^{-3} at that temperature. A) 1.71×10^4 mol dm^{-3} B) 1.71×10^{-4}mol dm^{-3} C) 2.71×10^{-4}mol dm^{-3} D) 3.42×10^{-5}mol dm^{-3}	1	4	2	3

THE HELPERS

9. If our eyes travel in counter clockwise direction from the ligand of highest priority to the ligand of lowest priority, the configuration is 1 2 3 2

A) S-Configuration B) R-Configuration
C) E-Configuration D) Z-Configuration

10. When a molecule has a plane of symmetry, it will be _____ 1 2 3 2

A) Optically active B) Both optically active and optically inactive
C) Enantiomer D) Optically inactive

11. The infinity of intermediate conformations are called 1 2 3 2

A) Staggered conformations B) Eclipsed conformations
C) Gauche D) Skew

12. Which step in S_N1 reaction is a slow rate determining step? 1 2 3 2

A) Attack of nucleophile B) Formation of transition state
C) Formation of racemic mixture D) both attack of nucleophile and formation of racemic mixture

13. Which of the following polymer type is not classified on the basis of its application and properties? 1 2 4 1

A) rubbers B) plastics
C) fibres D) synthetic

14. Which of the following monomers are unsuitable for condensation polymerization? 1 2 4 1

A) propanoic acid and ethanol B) butane-dioic acid and glycol
C) diamines and dicarboxylic acids D) hydroxy acids

15. The polymer in which steric placements of the substituent are arranged in such a way to give alternate d and 1 cinfigurations, is known as _____ 1 2 4 4

A) isotactic polymer B) atactic polymer
C) syndio-tactic polymer D) crosslinked polymers

16. Which of the following is a co-polymer? 1 2 4 1

A) Polythene B) Bakelite
C) PVC D) Polyacrylonitrile

17. In fiber reinforced composites longitudinal strength is mainly influenced by _____ 1 2 5 1

A) Fiber strength B) Fiber orientation
C) Fiber volume fraction D) Fiber length

18. In storage battery plates the matrix commonly used is _____ 1 2 5 1

A) Aluminium B) Lead
C) Silver D) Copper

19. Which of the following methods use soft X-rays to eject electrons from inner shell orbitals? 1 2 5 5

A) Auger electron spectroscopy B) Electron impact spectroscopy
C) X-ray crystallography D) X-ray photoelectron spectroscopy

20. If X-ray of wavelength 100 Å is incident on an atom at an angle of 90°, then what should be the value of d for first-order spectrum? 1 4 5 5

A) 30 Å B) 40 Å
C) 50 Å D) 60 Å

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PART - B (5 x 8 = 40 Marks)
Answer **ALL** the Questions

		Marks	BL	CO	PO
21 a.	i)Write a short note on spectrochemical series? Explain the impact of ligands in the formation of high and low spin complexes with suitable examples. [6] ii) What is Pairing energy (P)? Give the relation between crystal field splitting in octahedral complexes (Δ_O) and pairing energy (P). [2]	8	2	1	4
	(OR)				
b.	i) Write a note on the geometrical isomerism exhibited by octahedral complexes. ii) Aluminium occurs in nature as an oxide, whereas Platinum occurs as sulfide. Explain it on the basis of HSAB principle..	8	2	1	1
22 a.	Derive Nernst equation and give its application towards emf determination by potentiometric titration.	8	2	2	1
	(OR)				
b.	Explain in detail the mechanism with equations involved in dry corrosion.	8	2	2	2
23 a.	Sketch the potential diagram and discuss in detail the conformational analysis of n-butane.	8	3	3	3
	(OR)				
b.	Discuss elements of symmetry with an example for each.	8	3	3	4
24 a.	i) Give the differences between addition and condensation reactions. ii) Compare isotactic and syndiotactic polymers	8	1	4	1
	(OR)				
b.	Explain in detail n and p doping in conducting polymers.	8	1	4	1
25 a.	Explain the various regions in stress-strain graph with a neat diagram.	8	2	5	1
	(OR)				
b.	Discuss on ceramic and metal matrix composites with examples.	8	2	5	3

THE HELPERS

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Questions

		Marks	BL	CO	PO
26	Discuss in detail Pourbaix diagram of Iron system with a neat sketch.	15	4	2	3
27	a. What is the principle underlying in XPS? Give the merits and demerits of XPS analysis [8] b. What is Bragg's Law? Prove $n\lambda = 2d \sin\theta$. [7]	15	2	5	5

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